



FAIR 3D HERITAGE

*Documentation, Publication and Preservation
of the Digital 3D Heritage on the Web*

Completing the 3D Infrastructure: A One-Set Creation-to-Ingestion Model

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Keywords: 3D digitisation, mobile capture, FAIR 3D heritage, creation-to-ingestion workflow, institutional ownership

Abstract

Europe has made 3D digitisation a strategic priority for cultural heritage, supporting preservation, research, public access and reuse (European Commission, 2021). Initiatives such as Twin it! (European Commission, 2026) have translated this ambition into coordinated action. Yet despite growing technical capability, policy support and investment, production remains fragmented and limited in scale, and the wider vision of making large volumes of heritage assets available for meaningful and creative reuse remains unrealised.

This paper offers the perspective that this shortfall may arise not only from technological and funding constraints, but also from gaps in the sector's workflows and institutional structures.

The analysis first examines workflow. Conservation, research, archival documentation, public access and creative reuse place very different demands on accuracy, expertise, cost and scale. Discussing 3D digitisation as a single field can obscure the need for fit-for-purpose capture methods.

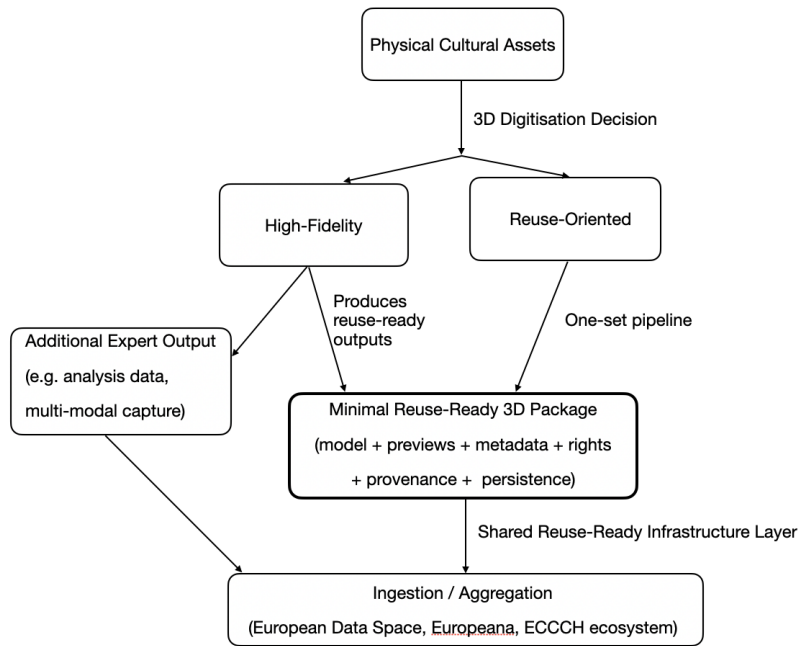


Fig. 1. Differentiated capture routes within the 3D heritage infrastructure. The upstream route for public-facing reuse across the long tail of collections remains underdeveloped.

Figure 1 highlights the resulting gap. High-fidelity capture is comparatively well established, while downstream initiatives such as 3DBigDataSpace (3DBigDataSpace Consortium, n.d.) are advancing the handling and integration of 3D heritage data. What remains underdeveloped is an upstream route for producing public-facing, reuse-ready 3D across the long tail of collections. The proposed model draws on observation of museum practice, analysis of existing workflows and infrastructure, and hands-on development and testing of mobile capture technologies.

For reuse at scale, capture must combine sufficient accuracy with ease of use, speed and affordability. Consumer mobile-capture applications can now produce high-quality models using established technologies, including photogrammetry, depth sensing and, increasingly, Gaussian splatting. They reconstruct recorded objects from captured data and do not use generative AI. A preliminary heritage study found that smart-device LiDAR models could be produced within minutes, with an average scale deviation of 1.89% from photogrammetric reference models (Hegarty & Saari, 2025). These workflows offer a practical route for trained non-specialists to produce reuse-oriented 3D at scale.

Yet a 3D model cannot travel through infrastructure alone. It must remain connected to institutional metadata, rights, provenance and stable access. I therefore propose a continuous creation-to-ingestion pathway coordinating capture, processing, institutional validation, packaging and downstream ingestion to produce an ingestion-ready unit by default.

Concretely, museum staff select an object; trained in-house staff undertake guided mobile capture; and automated processing is followed by institutional preview and approval. The system then generates a minimum reuse-ready package containing a web-ready model, preview assets, metadata linkage, rights and licensing information, paradata and a stable

access point. Together, these elements embed FAIR principles (Wilkinson et al., 2016) into production from the outset.

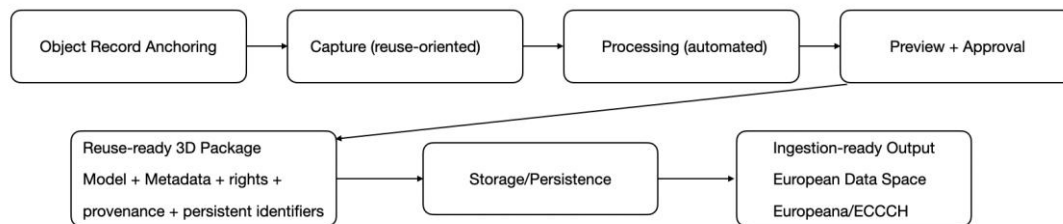


Fig. 2. The proposed one-set creation-to-ingestion workflow, connecting guided mobile capture, institutional validation and reuse-ready packaging to downstream infrastructure.

The second issue is the institutional structure needed to sustain reuse-oriented 3D digitisation. In many museums, 3D capture sits within collections teams focused on conservation and documentation, leaving no clear ownership of public-facing reuse. One possible structure is a hybrid model combining curator-led selection, trained in-house capture and collections-team support for handling, records and validation.

Europe's 3D ambition is achievable with today's technology. Realising it at scale will require fit-for-purpose capture, an end-to-end route from creation to ingestion, and clear institutional ownership of reuse-oriented digitisation - unlocking greater societal value from Europe's cultural heritage.

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